



University of Wisconsin
**SCHOOL OF MEDICINE
AND PUBLIC HEALTH**

Covariate-Adjusted Topological Inference and Learning

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Abstract: Many structured datasets—from neuroimaging to genomics—are influenced by nuisance covariates such as age, sex, or acquisition site. Traditional machine learning techniques for embedding and clustering often overlook these effects, leading to confounded comparisons where true biological differences are obscured by covariate-driven variation. We address this limitation by formulating covariate adjustment, including data harmonization, as defining equivalence classes in the sense of group theory: two data objects are equivalent if they differ only by the action of a covariate operator representing variation explained by covariates. The resulting latent quotient space provides a statistically principled framework for non-Euclidean and structured data, partitioning observations into equivalence classes that capture intrinsic structure after removing nuisance effects.

We demonstrate this framework in topological data analysis (TDA), where persistent topological features are embedded in a Hilbert space and adjusted via orthogonal projection, thereby constructing the quotient space. This enables downstream tasks such as embedding and clustering to be performed free from confounding influences. As a primary application, we analyze human brain networks from resting-state fMRI, adjusting for covariates such as sex, age, and acquisition site. Covariate-adjusted TDA markedly reduces variability and improves interpretability, offering a powerful, generalizable approach for reproducible and robust neuroscience.

Talk is based partially based on

Vector Space of Cycles

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Hernando Ombao²

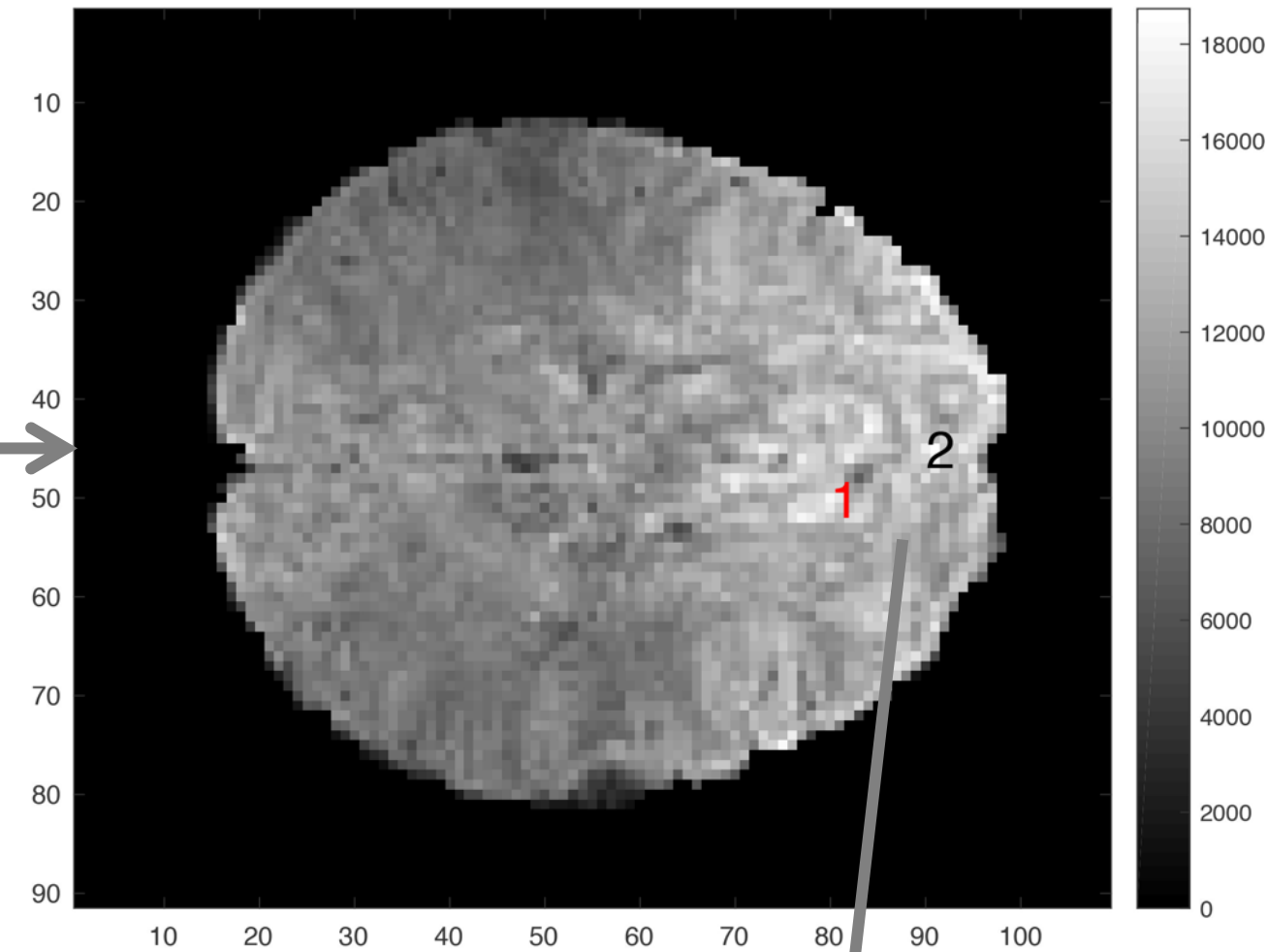
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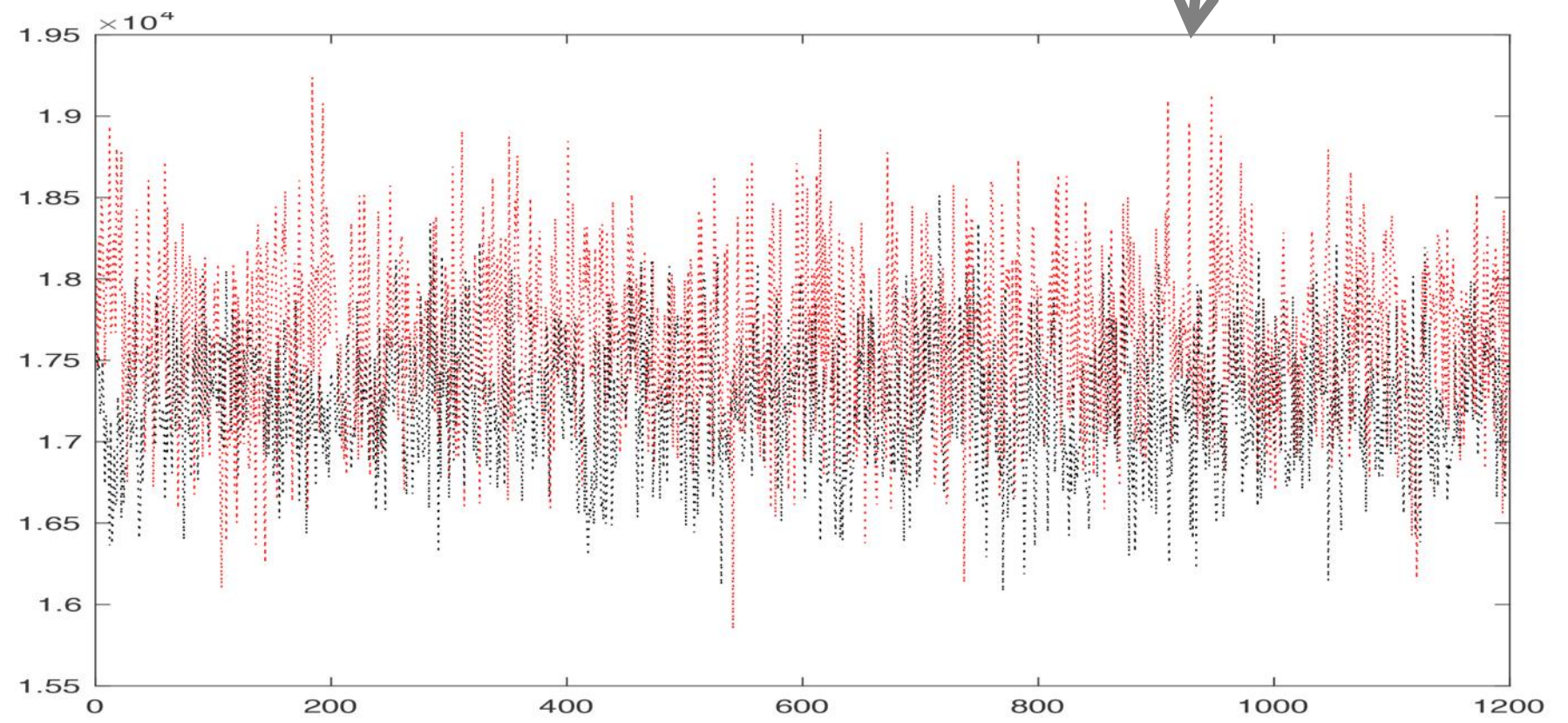
[arXiv:2606.08202](https://arxiv.org/abs/2606.08202)

Grants: NIH MH133614, NSF DMS-2010778

Resting-state functional magnetic resonance imaging (rs-fMRI)

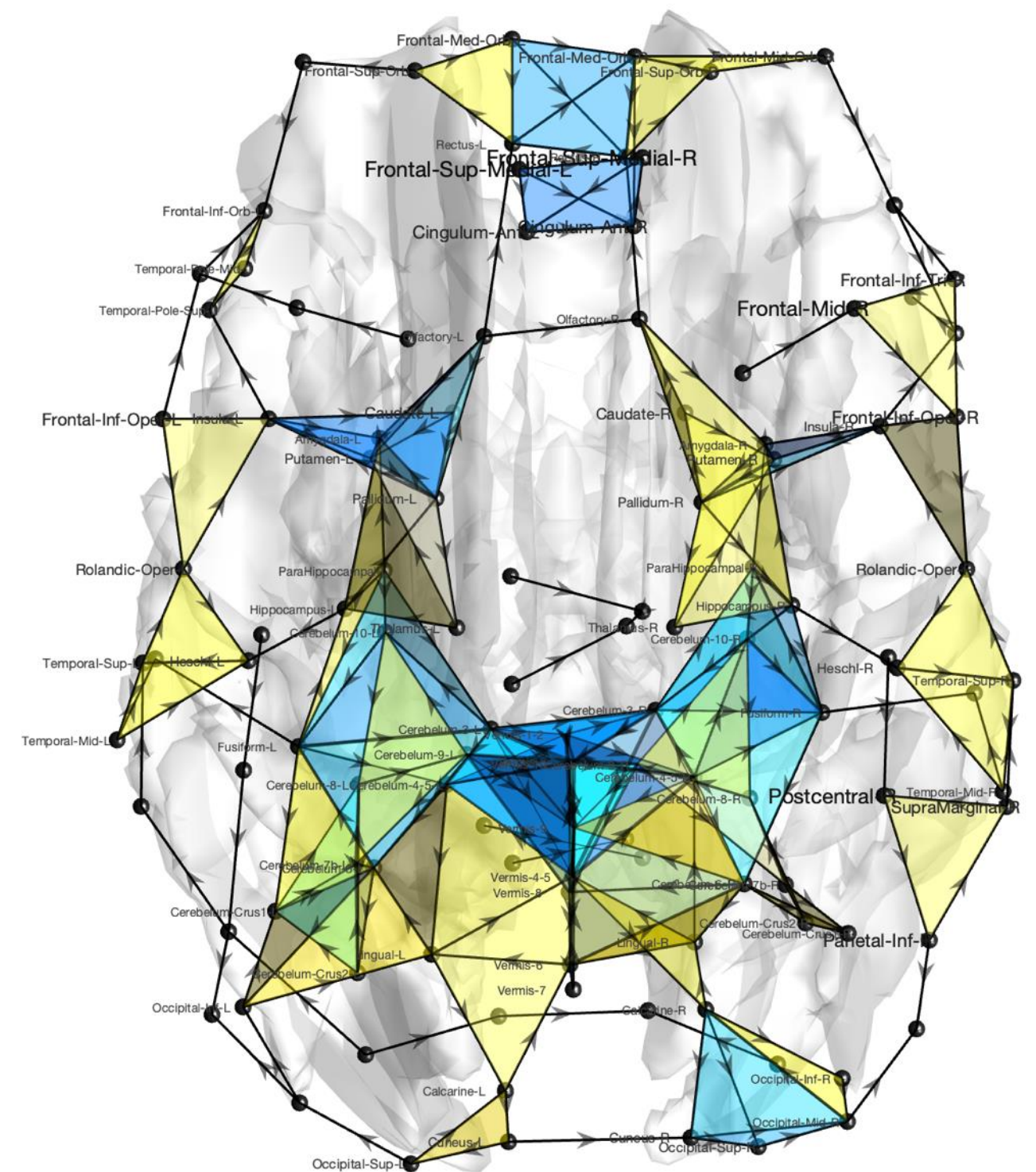
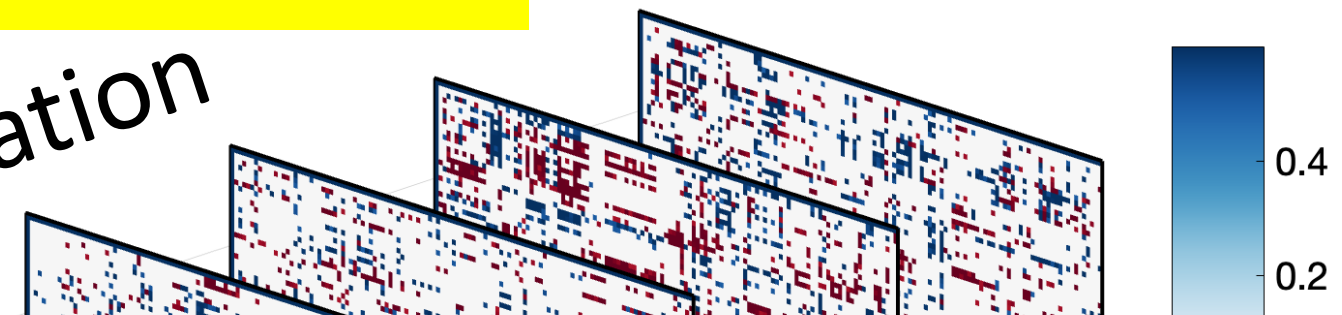
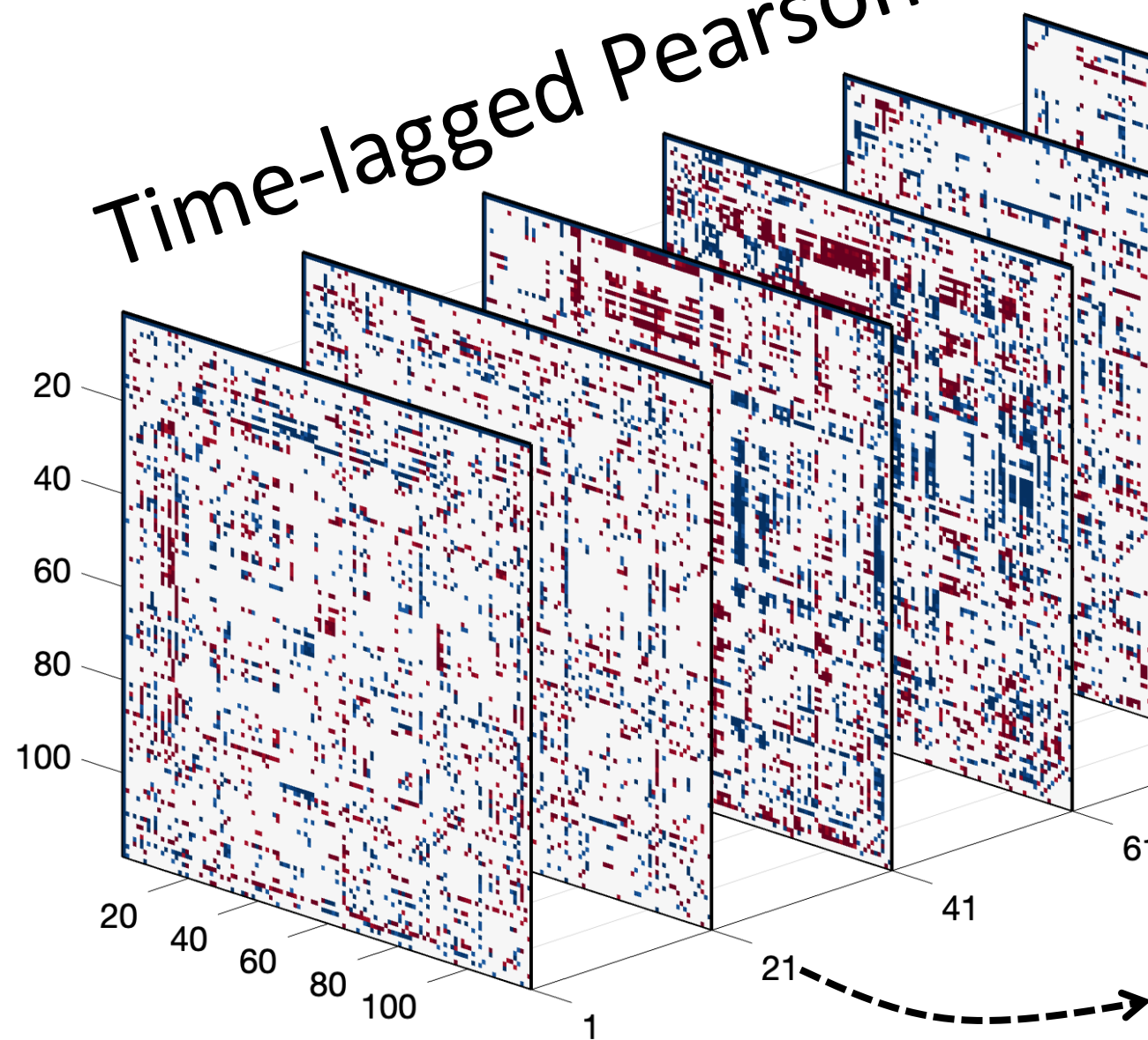


Neuronal activity over 14min.



Simplicial complex data structure

Time-lagged Pearson correlation

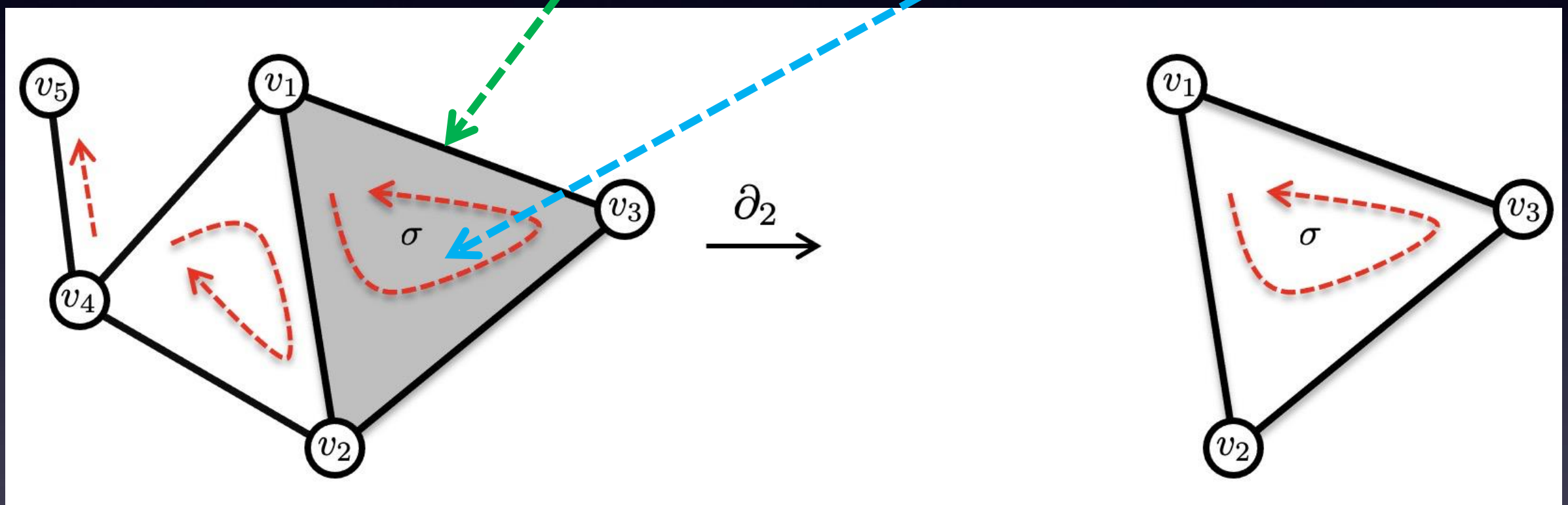


1st Hodge Laplacian

*Anand and Chung, 2023, IEEE
TMI*

Anand and Chung, 2024 ISBI

$$\Delta_1 = \overset{\substack{\text{\# of nodes} \\ \text{\# of edges}}}{\mathbf{B}_1^T \mathbf{B}_1} + \overset{\substack{\text{\# of edges} \\ \text{\# of faces}}}{\mathbf{B}_2 \mathbf{B}_2^T}$$



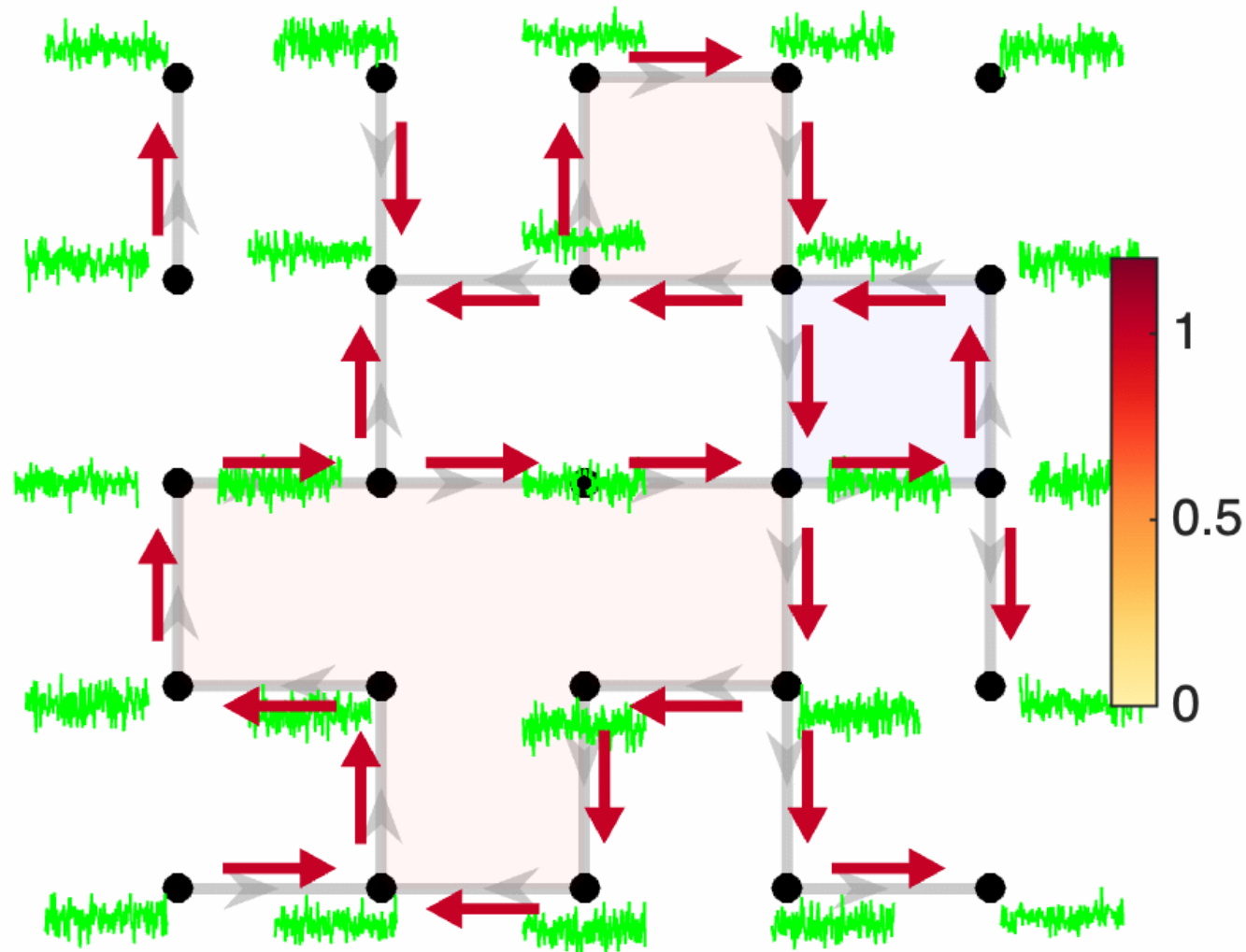
0th Hodge Laplacian
Graph Laplacian

$$\Delta_0 = \overset{\substack{\text{\# of edges} \\ \text{\# of nodes}}}{\mathbf{B}_1^T \mathbf{B}_1}$$

Dirichlet-Hodge diffusion Dynamics

Diffusion on edge flow

Damped Euler-Lagrangian (t = 0.00)



Chung et al. 2026 ISBI

$$\dot{X}(t) = -\Delta_1 X(t)$$

$$X(t) = e^{-\Delta_1 t} X(0)$$

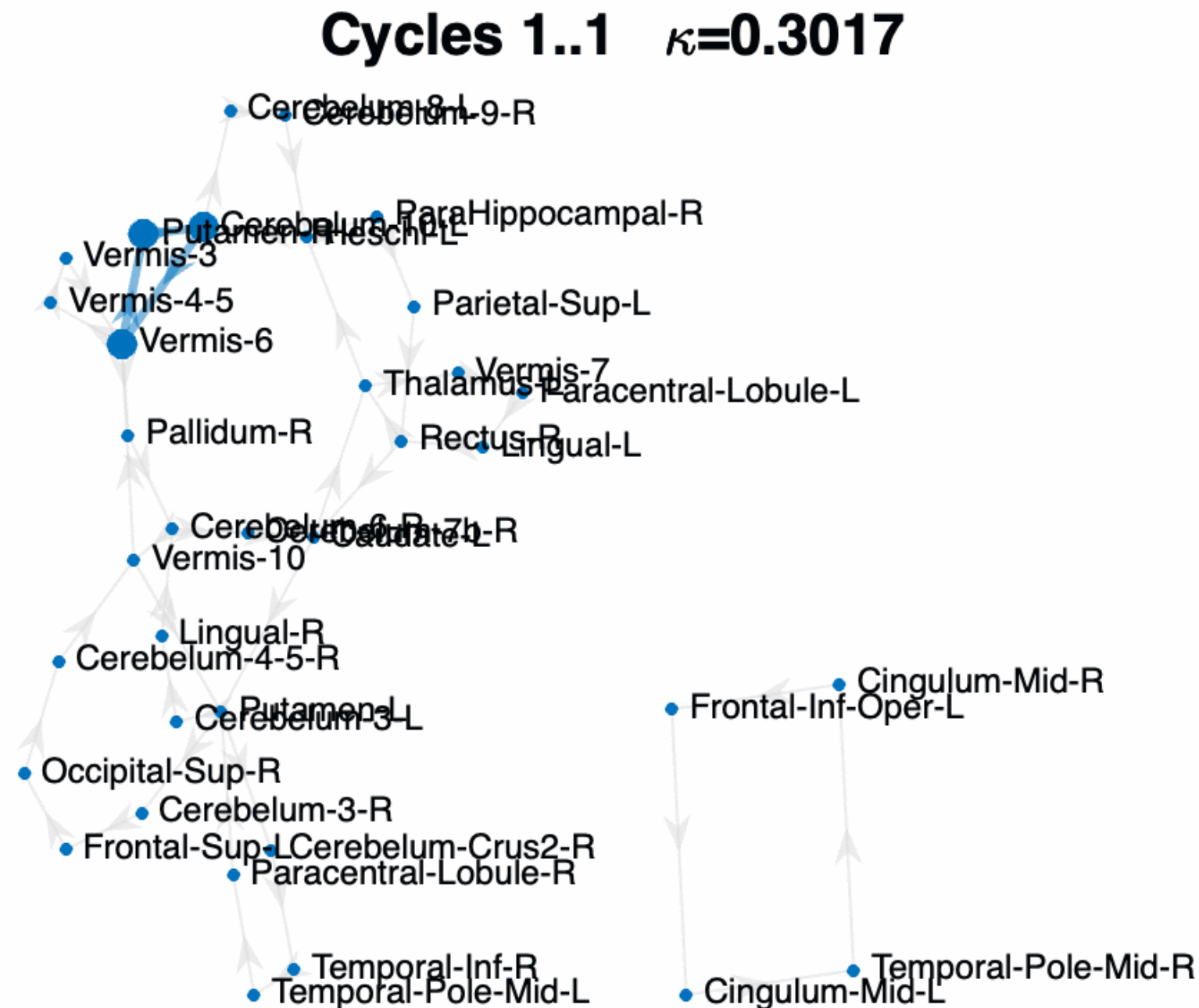
$$\sum_K e^{-\lambda_k t} \psi_k \psi_k^\top$$

Cyclic interactions

$$X_H = \mathcal{P}_H X(0)$$

$$\sum_k \psi_k \psi_k^\top$$

Average cyclic interactions of 400 subjects over 14min

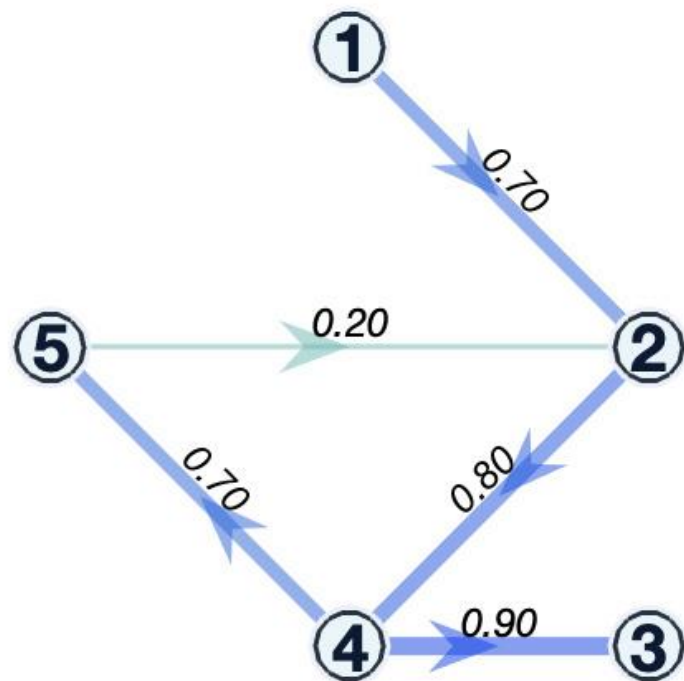


How to covariate adjust cyclic interactions
under the influence of sex, site, ages etc.

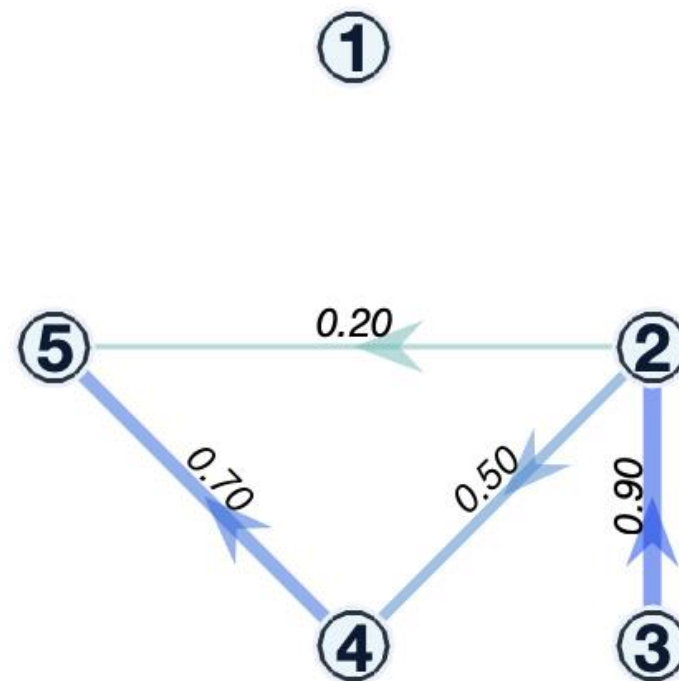
Covariating edges will destroy cycles.

$\ker(\Delta_1) \neq$ Vector space of cycles

Network 1



Network 2

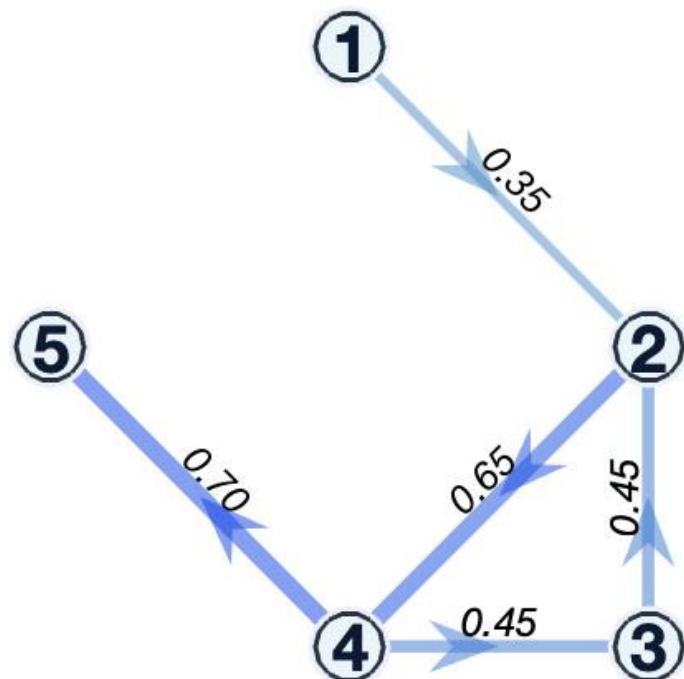


Cycles

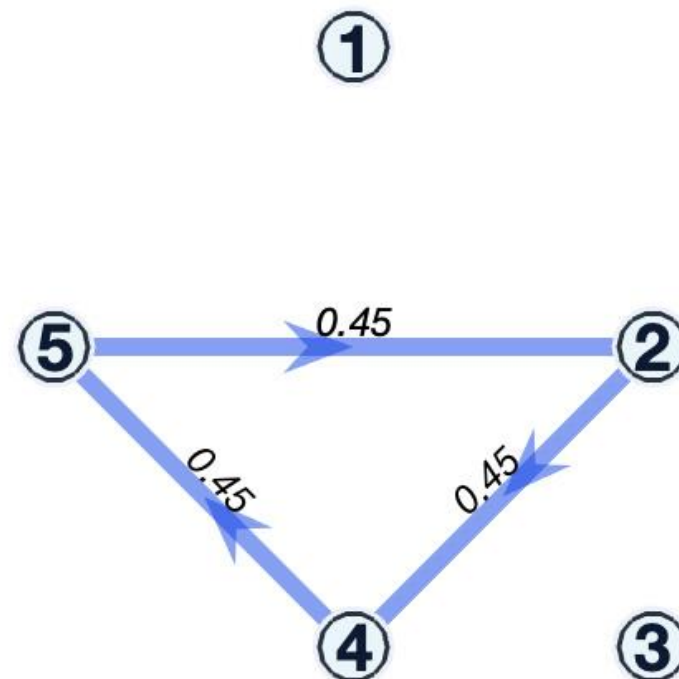
$$C_1, C_2 \in \ker(\Delta_1)$$

$$\alpha_1 C_1 + \alpha_2 C_2 \in \ker(\Delta_1)$$

Algebraic Averaging



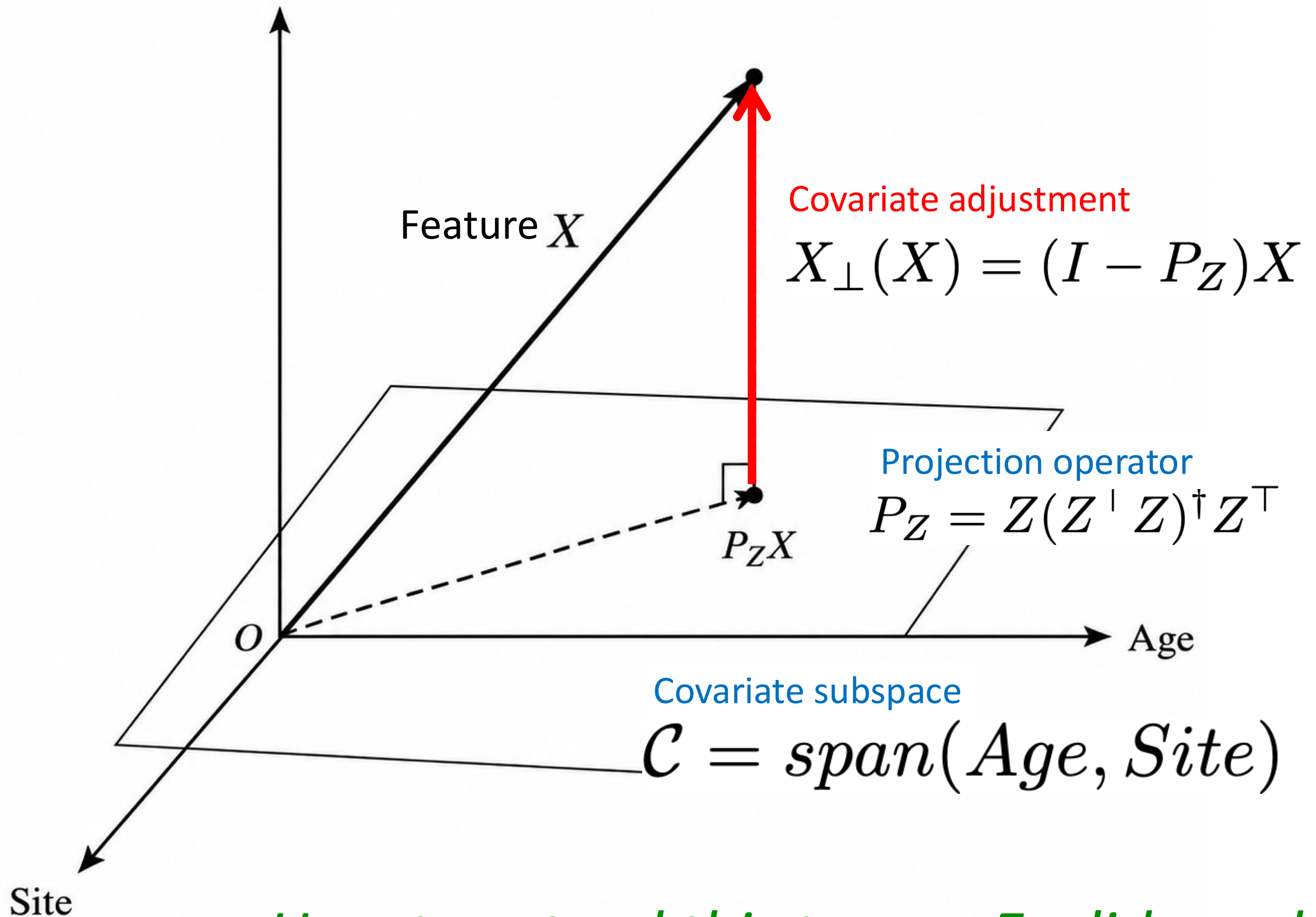
Averaging in Cycle Space



Anand and
Chung, 2023,
IEEE TMI

*Need to develop a
more general method*

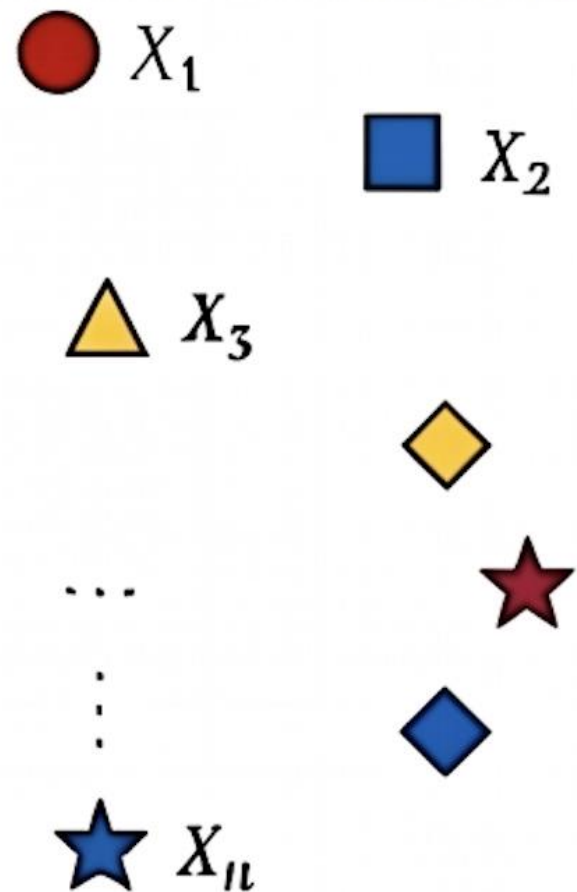
Covariate adjustment as a projection operator



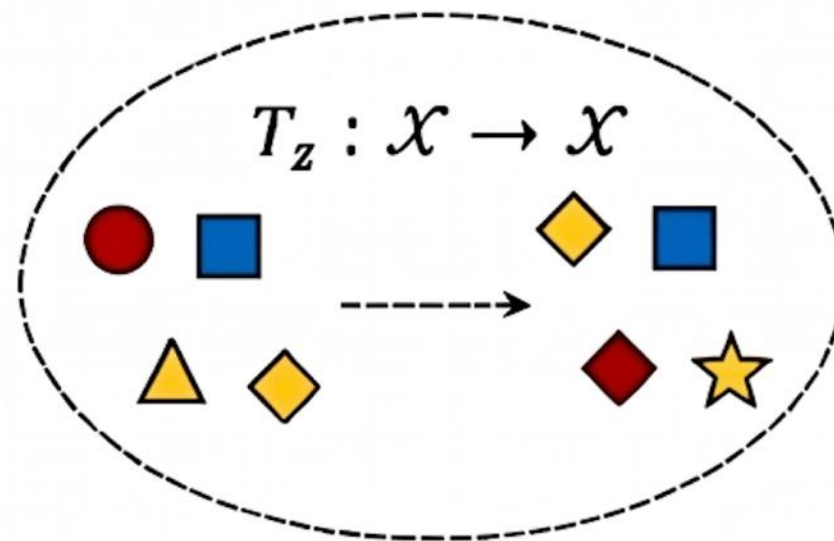
How to extend this to non-Euclidean data

Covariate adjustment via group action

Data Space $\mathcal{X}$



Affect of covariates

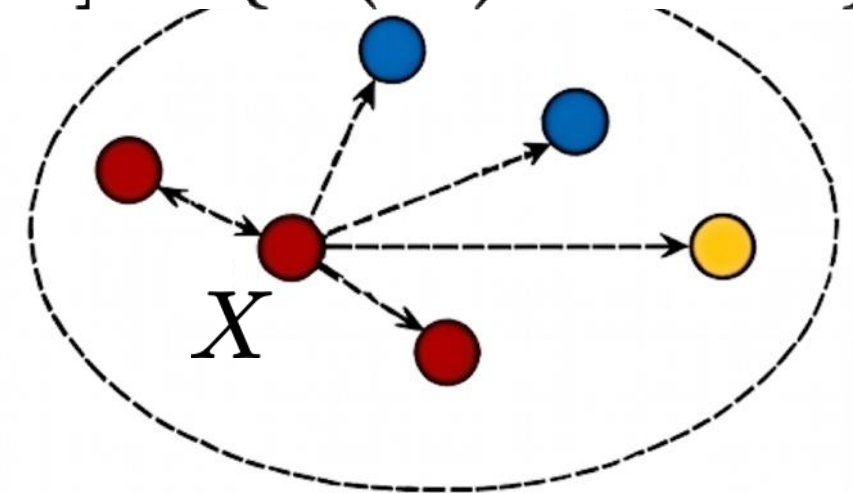


Group

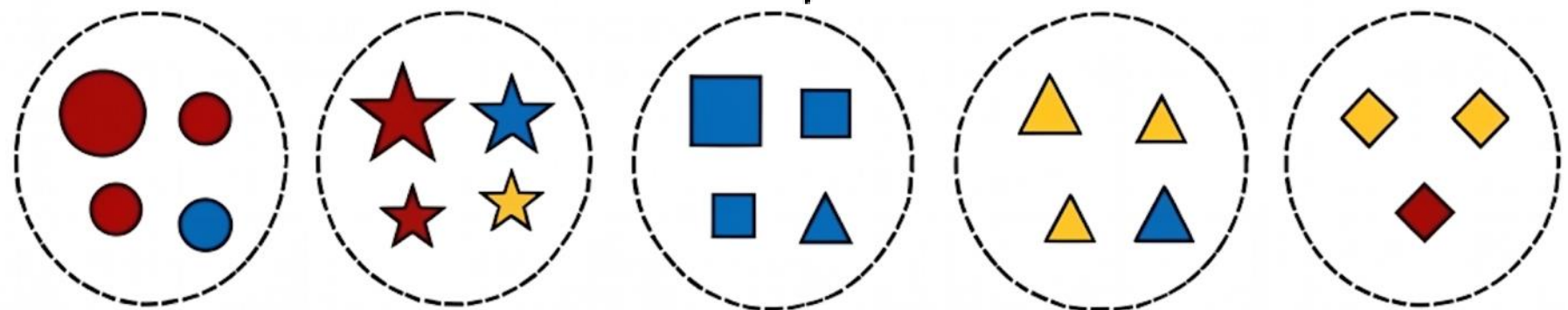
$$\mathcal{G} = \{T_z : z \in \text{span}(Z)\}$$

Equivalence class

$$[X] = \{T(X) : T \in \mathcal{G}\}$$

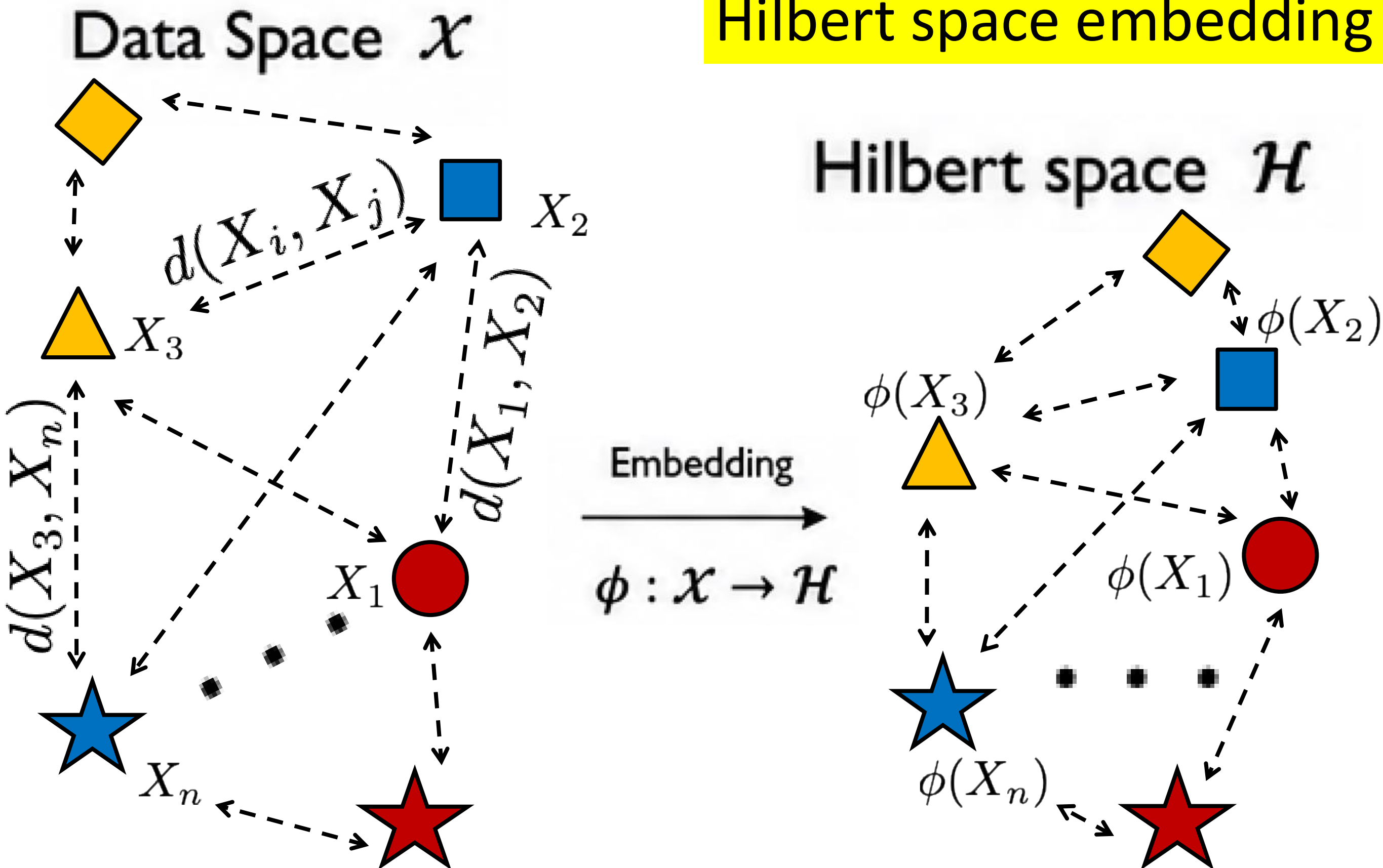


Quotient space $\mathcal{X}/\mathcal{G}$



How to compute these in

Hilbert space embedding



$$d(X_i, X_j) = \|\phi(X_i) - \phi(X_j)\|$$

Covariate adjustment in embedding space

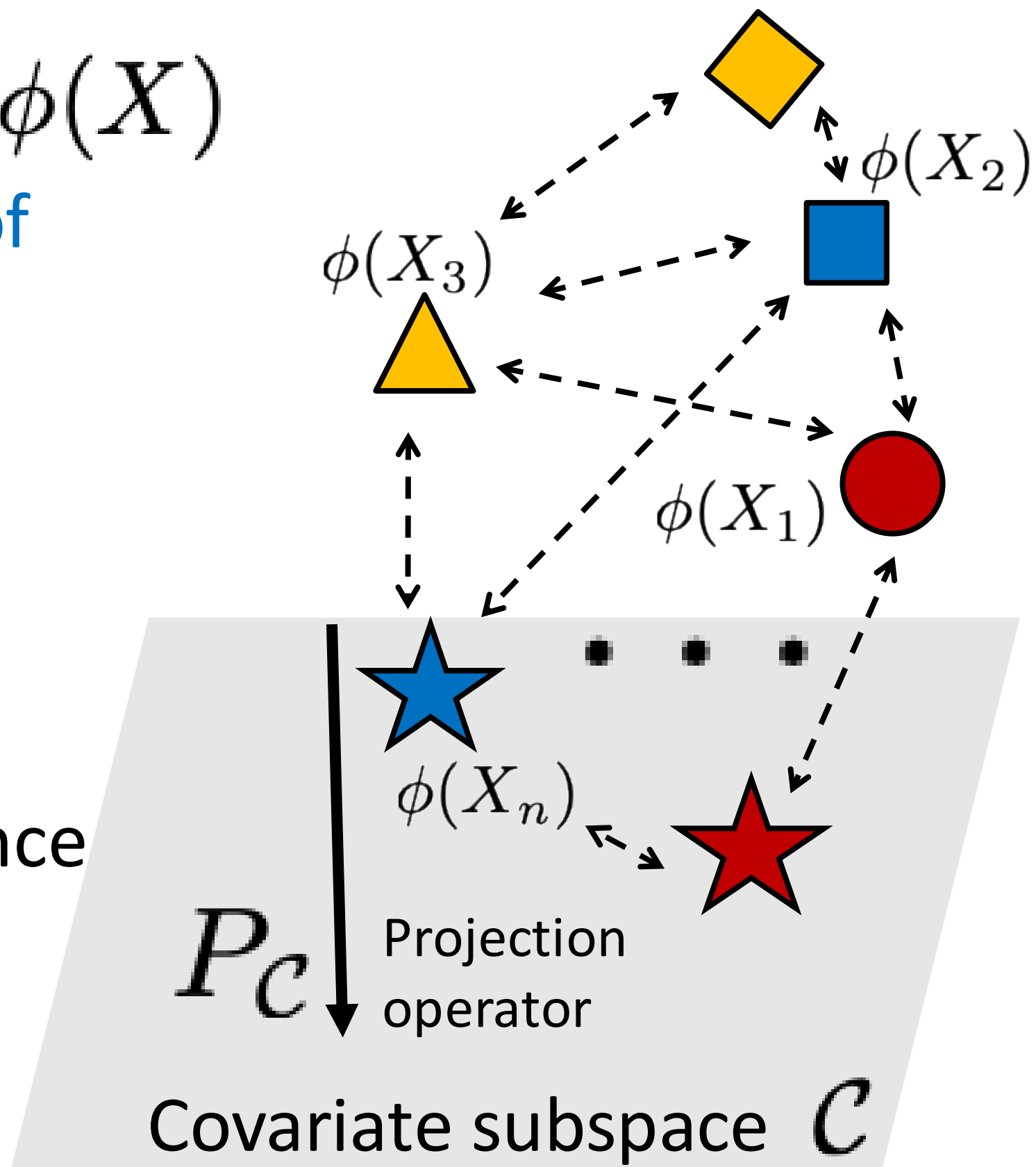
$$\phi_{\perp}(X) = (I - P_{\mathcal{C}})\phi(X)$$

Intrinsic representation of
object after removing
variation attributable to
nuisance covariates

Covariate adjusted distance

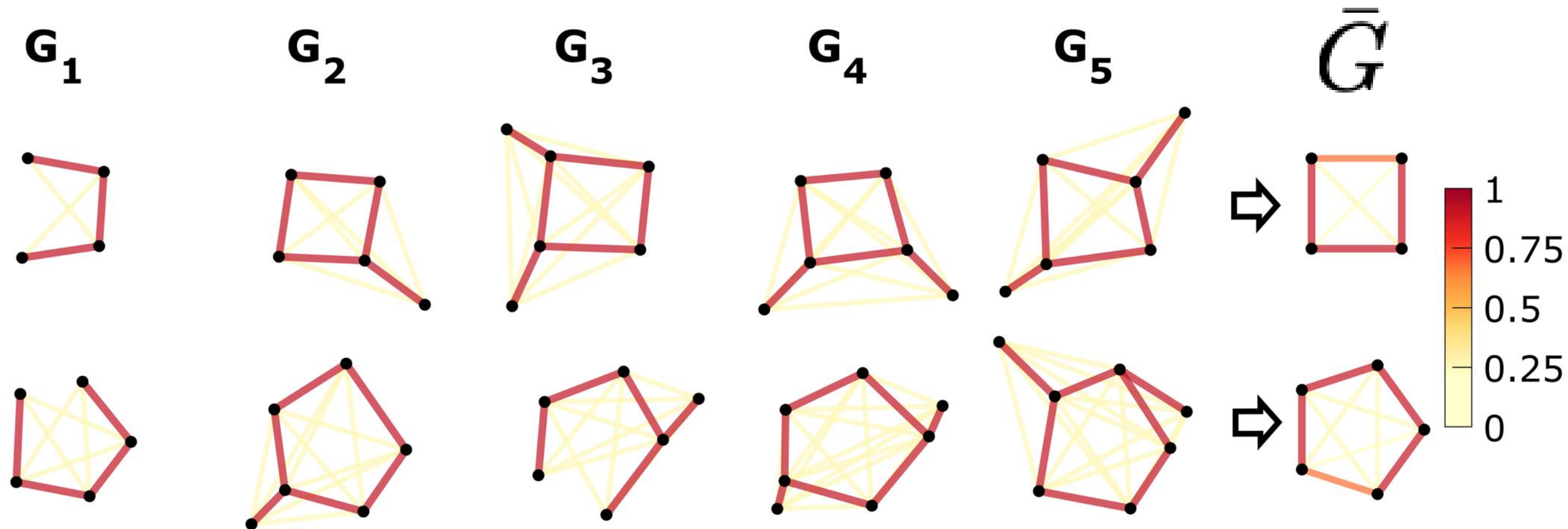
$$d_{\perp}(X_i, X_j) = \|\phi_{\perp}(X_i) - \phi_{\perp}(X_j)\|$$

Hilbert space $\mathcal{H}$



Application to topological distance

Wasserstein distance on persistent homology features



$$\bar{G} = \arg \min_{\Theta} \sum_{k=1}^n d(\Theta, G_i)$$

Analytic solution

Covariate adjusted topological clustering

ages

30

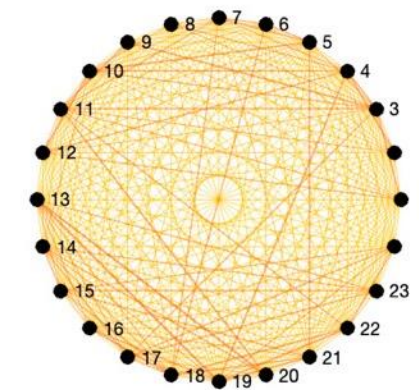
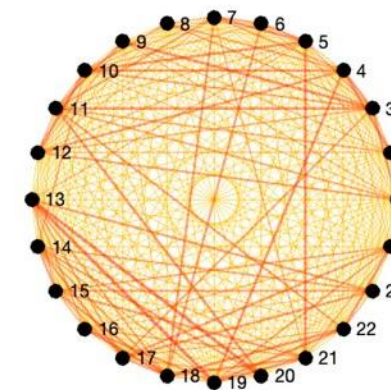
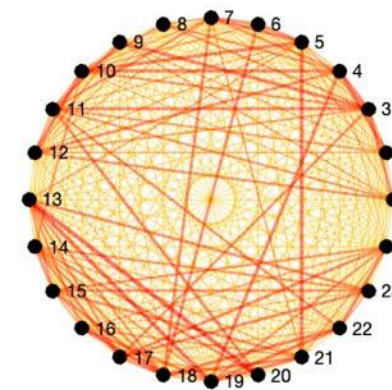
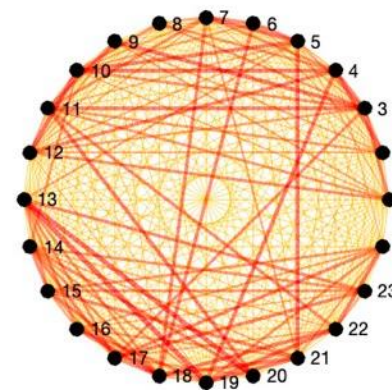
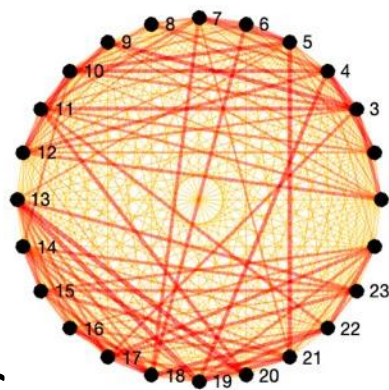
40

50

60

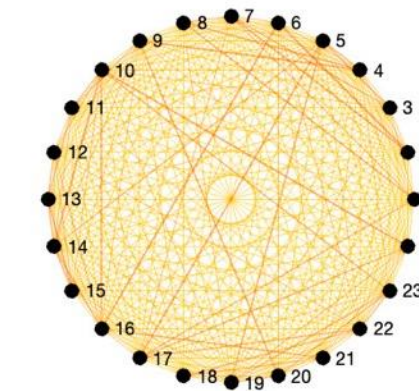
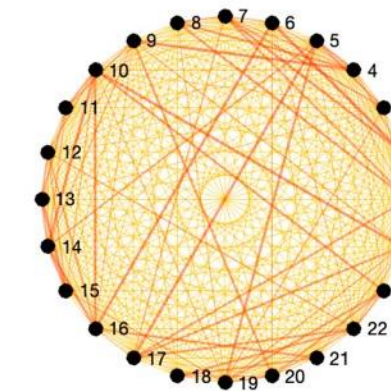
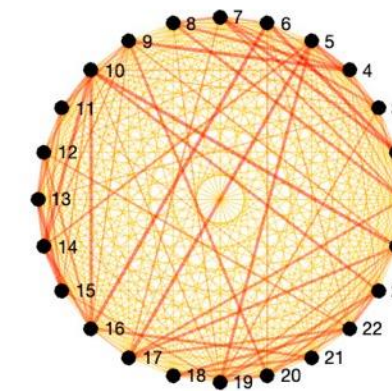
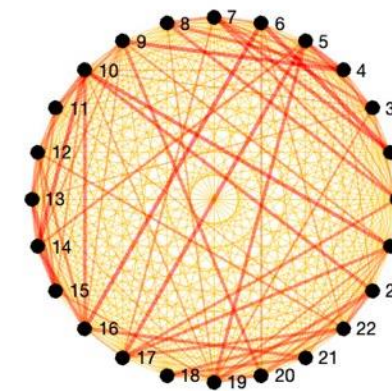
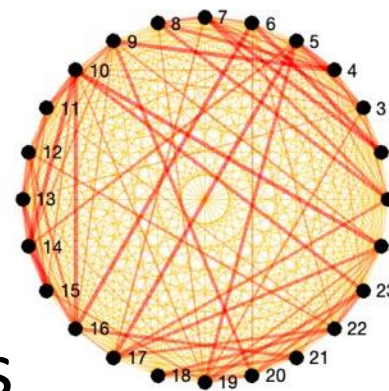
70

2-modules



Group 1

3-modules



Group 2

Connectivity



Accuracy

0.60+-0.02

After adjustment

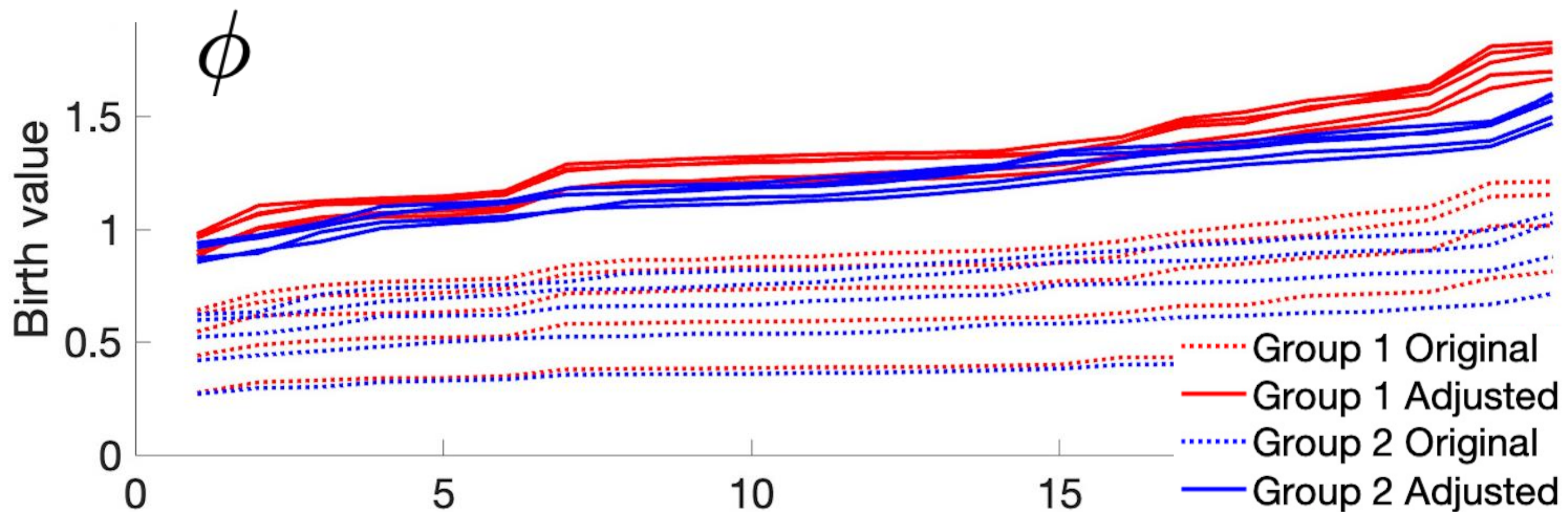
-----> 0.67+-0.05

Decomposition of Wassertein distance

$$d(X_i, X_j) = \|\phi(X_i) - \phi(X_j)\|$$

Wasserstein distance
on persistent homology
features

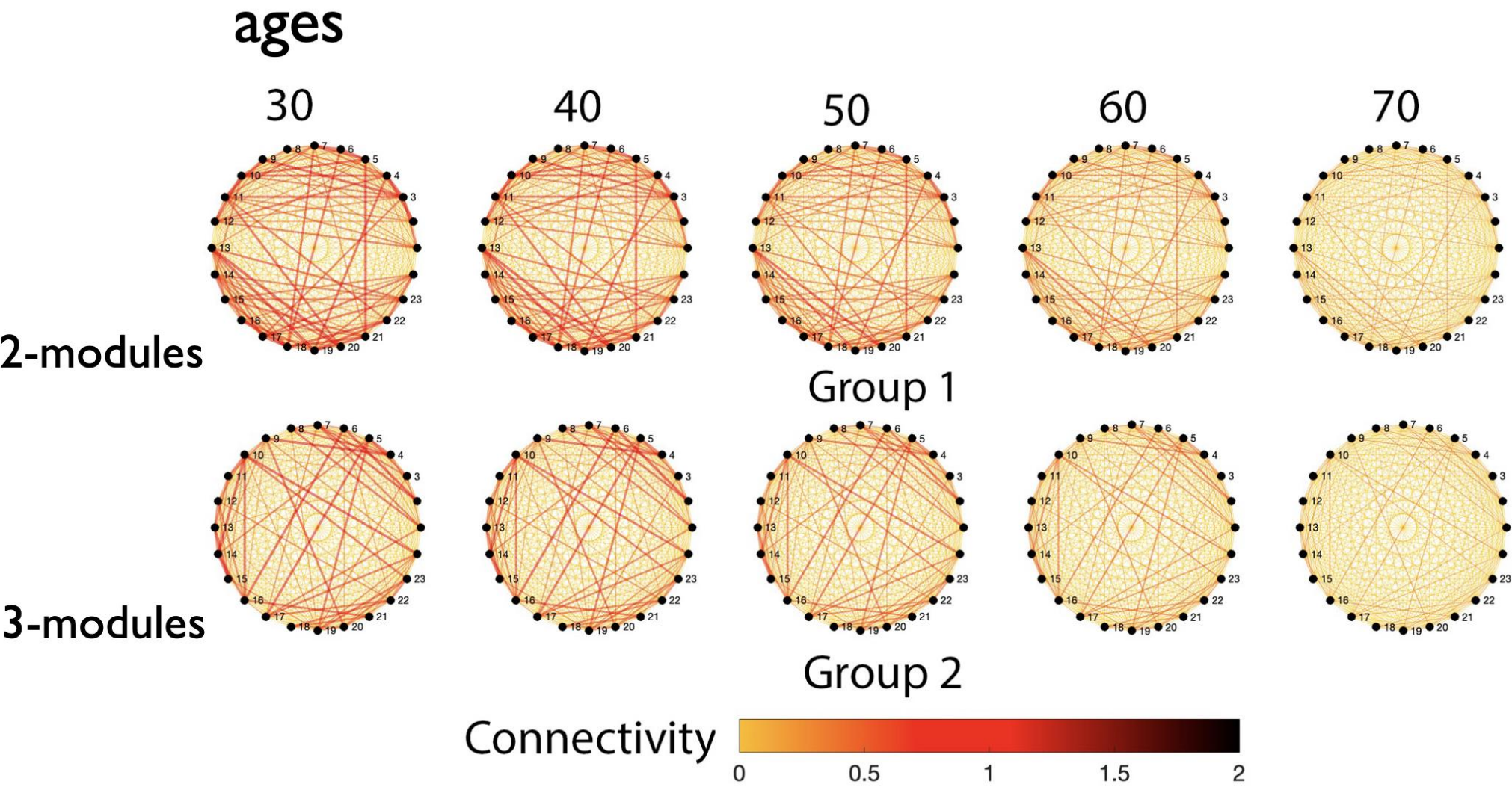
Monotone function



Need to solve projection with monotonicity

Null Test: False Positive Test

Important not to incorrectly cluster/classify in clinical setting



Method	Before Adjustment	After Adjustment	Error Reduction (%)
<i>k</i> -means (2 vs. 2 modules)	0.91 ± 0.09	0.66 ± 0.02	25%
Hierarchical (2 vs. 2 modules)	0.75	0.50	25%
<i>k</i> -means (3 vs. 3 modules)	0.90 ± 0.14	0.75 ± 0.04	15%
Hierarchical (3 vs. 3 modules)	0.75	0.75	0%

Thank you.

We are here.

